

JAVASCRIPT ASSIGNMENT : Arrays Advance Methods

SECTION 1 – map()

Q1. Square of Numbers

Given:

```
const numbers = [2, 4, 6, 8, 10];
```

Create a new array containing the square of every number.

Expected Output:

```
[4, 16, 36, 64, 100]
```

Q2. Convert Names to Uppercase

Given:

```
const names = ["nishant", "rahul", "amit", "priya"];
```

Convert every name to uppercase.

Expected Output:

```
["NISHANT", "RAHUL", "AMIT", "PRIYA"]
```

Q3. Add 10 to Every Number

Given:

```
const numbers = [10, 20, 30, 40, 50];
```

Add 10 to every element.

Expected Output:

```
[20, 30, 40, 50, 60]
```

Q4. Extract Employee Names

Given:

```
const employees = [  
  { name: "John", salary: 50000 },  
  { name: "David", salary: 60000 },
```

```
{ name: "Sarah", salary: 70000 }  
];
```

Use map() to create an array containing only employee names.

Expected Output:

```
["John", "David", "Sarah"]
```

SECTION 2 – filter()

Q5. Find Even Numbers

Given:

```
const numbers = [10, 15, 20, 25, 30, 35, 40];
```

Use filter() to return only even numbers.

Expected Output:

```
[10, 20, 30, 40]
```

Q6. Find Numbers Greater Than 50

Given:

```
const numbers = [25, 60, 45, 80, 30, 90, 55];
```

Return numbers greater than 50.

Expected Output:

```
[60, 80, 90, 55]
```

Q7. Filter Active Users

Given:

```
const users = [  
  { name: "John", active: true },  
  { name: "David", active: false },  
  { name: "Sarah", active: true },  
  { name: "Mike", active: false }  
];
```

Return only active users.

Expected Output:

```
[  
{ name: "John", active: true },  
{ name: "Sarah", active: true }  
]
```

Q8. Find Employees With Salary Greater Than 60,000

Given:

```
const employees = [  
  { name: "John", salary: 50000 },  
  { name: "David", salary: 75000 },  
  { name: "Sarah", salary: 90000 },  
  { name: "Mike", salary: 55000 }  
];
```

Return employees whose salary is greater than 60,000.

SECTION 3 – reduce()

Q9. Calculate Sum

Given:

```
const numbers = [10, 20, 30, 40, 50];
```

Use reduce() to calculate the total.

Expected Output:

150

Q10. Find Maximum Number

Given:

```
const numbers = [45, 78, 23, 90, 56, 12];
```

Use reduce() to find the maximum number.

Expected Output:

90

Q11. Calculate Total Salary

Given:

```
const employees = [  
  { name: "John", salary: 50000 },  
  { name: "David", salary: 60000 },  
  { name: "Sarah", salary: 70000 }  
];
```

Calculate the total salary of all employees.

Expected Output:

180000

Q12. Count Occurrences

Given:

```
const numbers = [1, 2, 2, 3, 1, 4, 2, 3, 3];
```

Use `reduce()` to count how many times each number occurs.

Expected Output:

```
{  
  1: 2,  
  2: 3,  
  3: 3,  
  4: 1  
}
```

SECTION 4 – Combining `map()` + `filter()`

Q13. Get Names of High-Salary Employees

Given:

```
const employees = [  
  { name: "John", salary: 50000 },  
  { name: "David", salary: 80000 },  
  { name: "Sarah", salary: 95000 },  
  { name: "Mike", salary: 45000 }  
];
```

Find employees whose salary is greater than 60,000 and return only their names.

Expected Output:

["David", "Sarah"]

Q14. Double Only Even Numbers

Given:

```
const numbers = [1, 2, 3, 4, 5, 6, 7, 8];
```

First filter the even numbers and then double them.

Expected Output:

[4, 8, 12, 16]

SECTION 5 – Combining filter() + reduce()

Q15. Total Salary of Employees With Salary > 60,000

Given:

```
const employees = [  
  { name: "John", salary: 50000 },  
  { name: "David", salary: 80000 },  
  { name: "Sarah", salary: 95000 },  
  { name: "Mike", salary: 45000 }  
];
```

Find the total salary of employees whose salary is greater than 60,000.

Expected Output:

175000

Q16. Sum of Even Numbers

Given:

```
const numbers = [10, 15, 20, 25, 30, 35, 40];
```

Use filter() and reduce() to calculate the sum of even numbers.

Expected Output:

100

SECTION 6 – INTERVIEW-LEVEL QUESTIONS

Q17. Product Price Calculation

Given:

```
const products = [  
  { name: "Laptop", price: 80000 },  
  { name: "Mouse", price: 1500 },  
  { name: "Keyboard", price: 3000 },  
  { name: "Monitor", price: 20000 }  
];
```

Calculate the total price of all products.

Q18. Find Highest-Paid Employee

Given:

```
const employees = [  
  { name: "John", salary: 55000 },  
  { name: "David", salary: 85000 },  
  { name: "Sarah", salary: 95000 },  
  { name: "Mike", salary: 65000 }  
];
```

Use `reduce()` to find the employee with the highest salary.

Expected Output:

```
{ name: "Sarah", salary: 95000 }
```

Q19. Calculate Average Salary

Given:

```
const employees = [  
  { name: "John", salary: 50000 },  
  { name: "David", salary: 70000 },  
  { name: "Sarah", salary: 90000 },  
  { name: "Mike", salary: 60000 }  
];
```

Calculate the average salary using `reduce()`.

Expected Output:

67500

Q20. IT Department Total Salary

Given:

```
const employees = [  
  { name: "John", department: "IT", salary: 60000 },  
  { name: "David", department: "HR", salary: 50000 },  
  { name: "Sarah", department: "IT", salary: 90000 },  
  { name: "Mike", department: "HR", salary: 70000 },  
  { name: "Alex", department: "IT", salary: 80000 }  
];
```

Find the total salary of IT employees using
filter() + reduce().

Expected Output:

230000

BONUS CHALLENGE

Solve the following problems using only:

map()

filter()

reduce()

Do NOT use traditional for/while loops.

Given:

```
const employees = [  
  { name: "John", age: 28, department: "IT", salary: 60000 },  
  { name: "David", age: 35, department: "HR", salary: 50000 },  
  { name: "Sarah", age: 30, department: "IT", salary: 90000 },  
  { name: "Mike", age: 25, department: "Finance", salary: 70000 },  
  { name: "Alex", age: 32, department: "IT", salary: 80000 }  
];
```

Task A:

Get names of all IT employees.

Task B:

Get names of employees whose salary is greater than 70,000.

Task C:

Calculate total salary of IT employees.

Task D:

Find the highest-paid employee.

Task E:

Calculate the average salary of all employees.

Task F:

Create an array containing employee names in uppercase.

Task G:

Find employees whose age is greater than 30
and return only their names.

Task H:

Find the total salary of employees whose age is greater than 30.